

1,004

THE NUMBER OF
STARS CATALOGUED
IN **TYCHO BRAHE'S**
*ASTRONOMIAE
INSTAURATAE
MECHANICA*

IN 1598, DANISH ASTRONOMER TYCHO BRAHE published *Astronomiae Instauratae Mechanica* (*Instruments for the Restoration of Astronomy*)—a star catalog listing **the positions of more than a thousand stars** he had observed. Brahe had recently left his observatories at Hven, Denmark, after falling out with his patron, King Christian IV, who did not share his predecessor's enthusiasm for astronomy. Before leaving, Brahe wrote detailed descriptions with illustrations of the instruments and equipment he had used at his observatories. He included these in *Astronomiae* as well. In the following year,

Brahe found a new sponsor, the Holy Roman Emperor Rudolf II, and moved to Prague (now in the Czech Republic).

In September, Dutch sailors landed on the island of Mauritius in the Indian Ocean, claiming it for the Netherlands. They were **the first to describe the dodo**, a flightless bird related to the pigeon and unique to Mauritius. Within less than a century, the bird had been **driven to extinction**, hunted by the

settlers and preyed upon by imported animals.

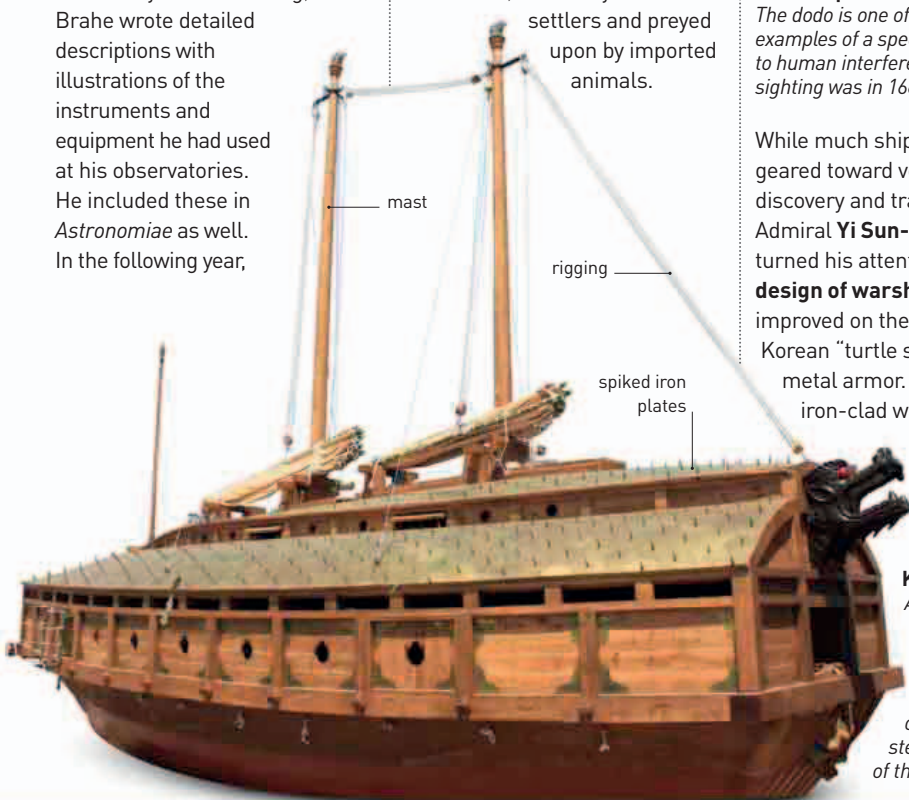


Extinct species

The dodo is one of the first recorded examples of a species' extinction due to human interference. Its last known sighting was in 1662.

While much ship design was geared toward voyages of discovery and trade, in 1598, Admiral **Yi Sun-sin** of Korea turned his attention to the **design of warships**. He improved on the traditional Korean "turtle ship" by adding metal armor. These first iron-clad warships were protected by iron plates covered with spikes.

Korean ship
Admiral Yi Sun-sin's redesign of the Korean "turtle ship" was a precursor of the iron-clad steam warships of the 19th century.



Giordano Bruno, arrested by the Roman Inquisition on charges of heresy, was imprisoned from 1562 until his execution in 1600.

ITALIAN NATURALIST ULISSE ALDROVANDI (1522–1605), who had founded the botanical garden of Bologna in 1568, published the first of three volumes of *Ornithologiae*—a treatise on birds—in 1599. As well as designing and managing the botanical garden, Aldrovandi organized expeditions to collect plants for his herbarium and established a large collection of flora and fauna specimens. He also wrote many books covering all aspects of natural history, helping to lay **the foundations for the modern study of botany and zoology**.

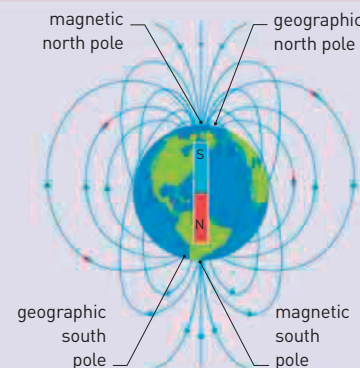
In 1600, English physician and scientist **William Gilbert** (1544–1603) published *De Magnete, Magneticisque Corporibus, et de Magno Magnete Tellure* (*On the Magnet, Magnetic*

31–34
MILES / YEAR
THE SPEED AT WHICH THE MAGNETIC NORTH POLE IS MOVING

Bodies, and the Great Magnet of the Earth). In this treatise, he described his experiments with magnets, many of which used small magnetic spheres called "terrellae," to model Earth's behavior. He concluded that **Earth behaves as a giant magnet**, making compass needles point north, and that **the center of Earth is made of iron**.

EARTH'S MAGNETIC POLES

The Earth's core—composed of an iron alloy—behaves like a gigantic bar magnet. Magnetic compass needles are attracted to the two poles of the Earth's magnetic core, also known as the magnetic poles. These coincide roughly with the geographical north and south poles. Because the core is fluid, the magnetic poles can shift position.



Tycho Brahe publishes the star catalog *Astronomiae Instauratae Mechanica*

Korean Admiral Yi Sun-sin develops an improved "turtle ship"

1599 Italian naturalist **Ulisse Aldrovandi** publishes the first volume of *Ornithologiae*

1599 Tycho Brahe builds an observatory at Benátky nad Jizerou

Dutch sailors discover the dodo on the island of Mauritius

1600 William Gilbert publishes *De Magnete, Magneticisque Corporibus, et de Magno Magnete Tellure*